1.08 Graph Sketching: Quadratic Graphs

Worked Solutions

1 a)

$y = (x + 3)(x - 3) \Rightarrow y = x^2 - 9$

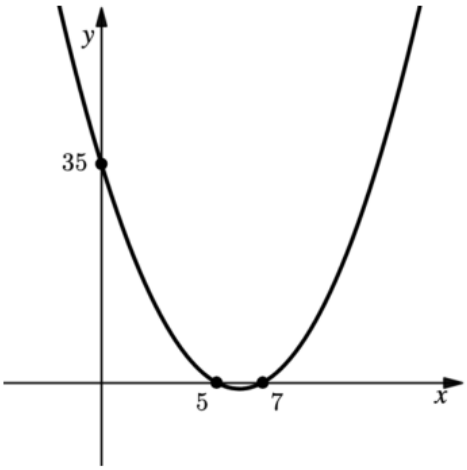
1 b)

$y = (x + 2)(x - 4) \Rightarrow y = x^2 - 2x - 8$

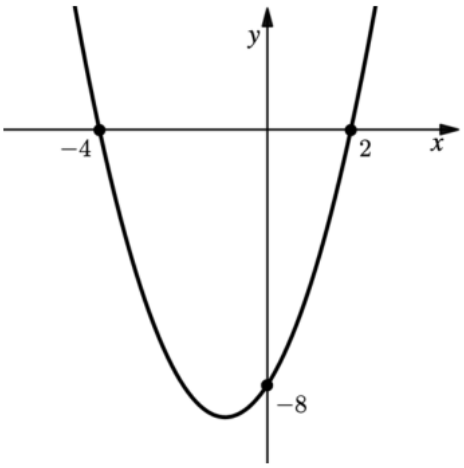
1 c)

$y = (x + 3)(x + 1) \Rightarrow y = x^2 + 4x + 3$

2 a)



2 b)



3 a)

$y = -(x+4)(x-4) \Rightarrow y = -x^2 + 16$

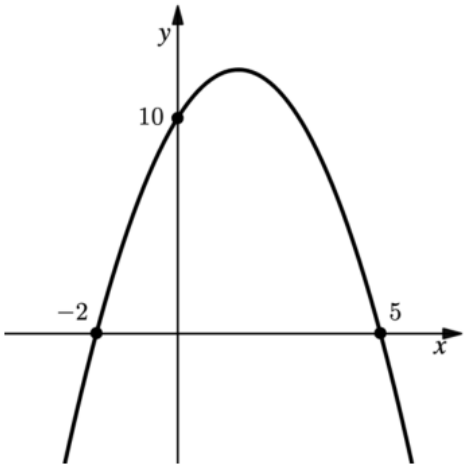
3 b)

$y = -(x+5)(x-1) \Rightarrow y = -x^2 - 4x + 5$

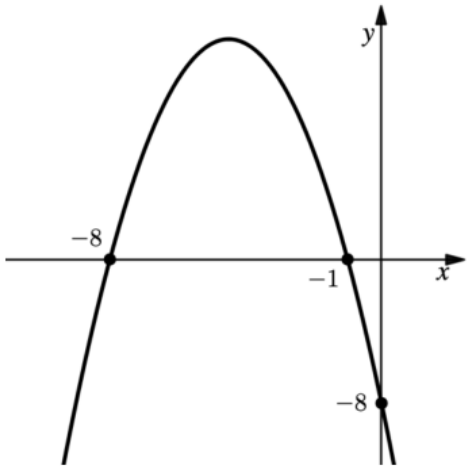
3 c)

$y = -(x-2)(x-6) \Rightarrow y = -x^2 + 8x - 12$

4 a)



4 b)



5 a)

$y = (x-2)^2 + 1 \Rightarrow y = x^2 - 4x + 5$

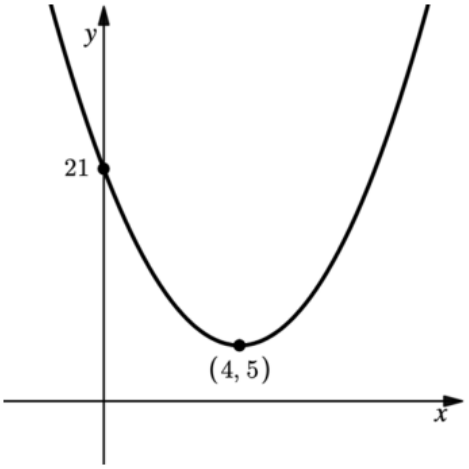
5 b)

$y = (x+2)^2 + 3 \Rightarrow y = x^2 + 4x + 7$

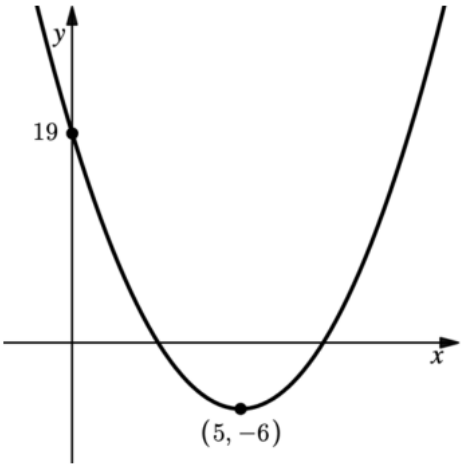
5 c)

$y = (x+3)^2 - 5 \Rightarrow y = x^2 + 6x + 4$

6 a)



6 b)



7 a)

$y = -(x-2)^2 - 3 \Rightarrow y = -x^2 + 4x - 7$

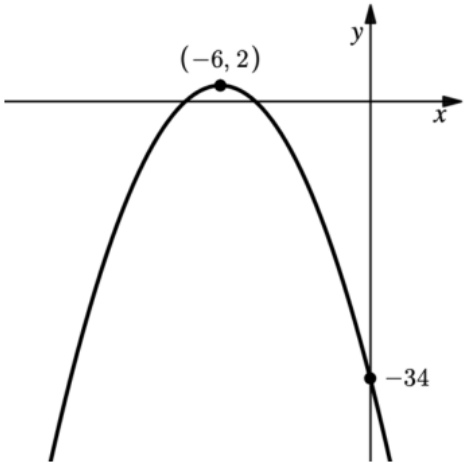
7 b)

$y = -(x-4)^2 + 3 \Rightarrow y = -x^2 + 8x - 13$

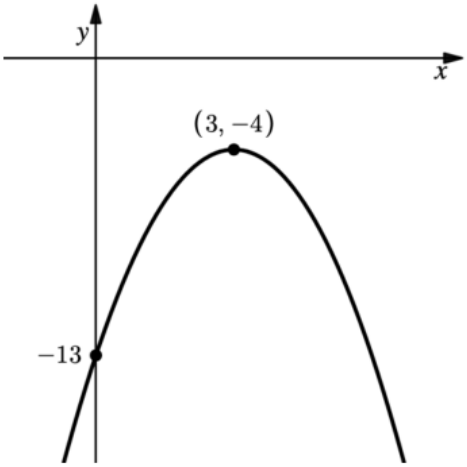
7 c)

$y = -(x+2)^2 + 6 \Rightarrow y = -x^2 - 4x + 2$

8 a)



8 b)



9 a)

$y = k(x+3)(x-2)$

$(0,-12) : -12 = k(3)(-2) \Rightarrow k = 2$

$y = 2(x+3)(x-2) \Rightarrow y = 2x^2 + 2x - 12$

9 b)

$y = k(x+4)(x+1)$

$(0,16) : 16 = k(4)(1) \Rightarrow k = 4$

$y = 4(x+4)(x+1) \Rightarrow y = 4x^2 + 20x + 16$

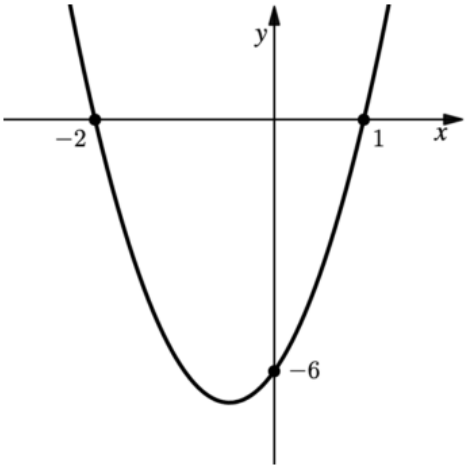
9 c)

$y = k(x+3)(x-2)$

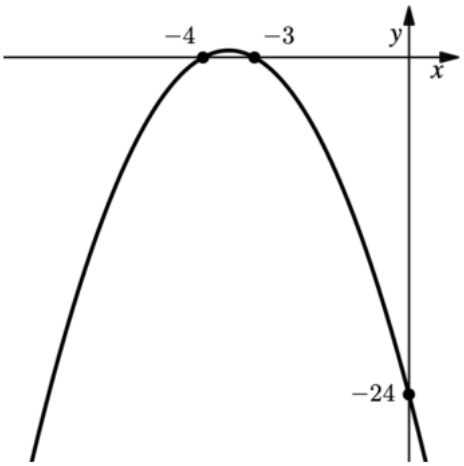
$(0,9) : 9 = k(3)(-2) \Rightarrow k = -\frac{3}{2}$

$y = -\frac{3}{2}(x+3)(x-2) \Rightarrow y = -\frac{3}{2}x^2 - \frac{3}{2}x + 9$

10 a)



10 b)



11 a)

$y = k(x-3)^2 + 4$

$(0, 31) : 31 = k(-3)^2 + 4 \Rightarrow k = 3$

$y = 3(x-3)^2 + 4 \Rightarrow y = 3x^2 - 18x + 31$

11 b)

$y = k(x+6)^2 - 5$

$(0, 7) : 7 = k(6)^2 - 5 \Rightarrow k = \frac{1}{3}$

$y = \frac{1}{3}(x+6)^2 - 5 \Rightarrow y = \frac{1}{3}x^2 + 4x + 7$

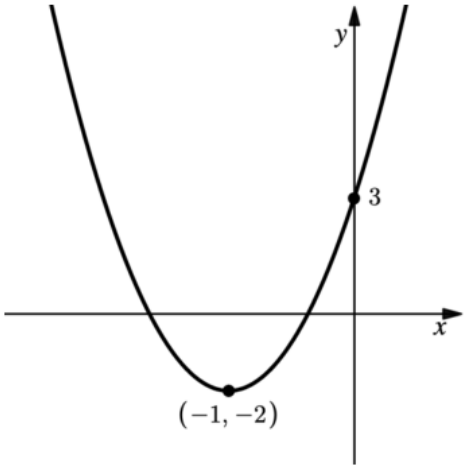
11 c)

$y = k(x-4)^2 + 4$

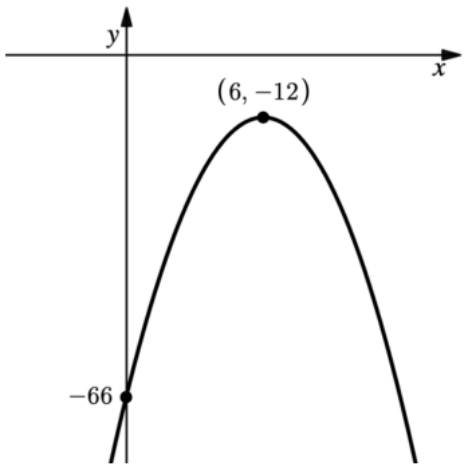
$(0, -16) : -16 = k(-4)^2 + 4 \Rightarrow k = -\frac{5}{4}$

$y = -\frac{5}{4}(x-4)^2 + 4 \Rightarrow y = -\frac{5}{4}x^2 + 10x - 16$

12 a)



12 b)



13 a)

$y = k(x - (1 - \sqrt{3}))(x - (1 + \sqrt{3}))$

$(0, -4) : -4 = k(-(1 - \sqrt{3}))(-(1 + \sqrt{3})) \Rightarrow k = 2$

$y = 2(x - (1 - \sqrt{3}))(x - (1 + \sqrt{3}))$

$\Rightarrow y = 2x^2 - 4x - 4$

13 b)

$y = k(x - (-4 - \sqrt{6}))(x - (-4 + \sqrt{6}))$

$(0, -25) : -25 = k(-(-4 - \sqrt{6}))(-(-4 + \sqrt{6}))$

$\Rightarrow k = -\frac{5}{2}$

$y = -\frac{5}{2}(x - (-4 - \sqrt{6}))(x - (-4 + \sqrt{6}))$

$\Rightarrow y = -\frac{5}{2}x^2 - 20x - 25$

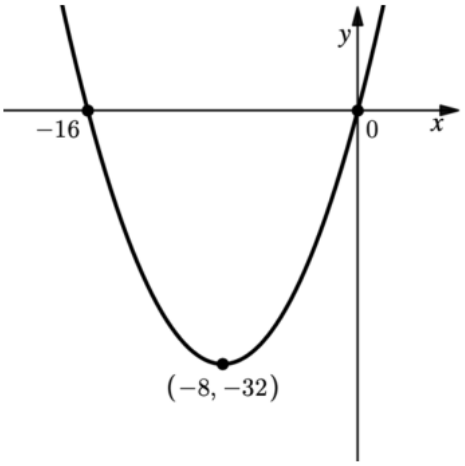
13 c)

$y = k(x-4)^2 + 12$

$(0,0) : 0 = k(-4)^2 + 12 \Rightarrow k = -\frac{3}{4}$

$y = -\frac{3}{4}(x-4)^2 + 12 \Rightarrow y = -\frac{3}{4}x^2 + 6x$

14)

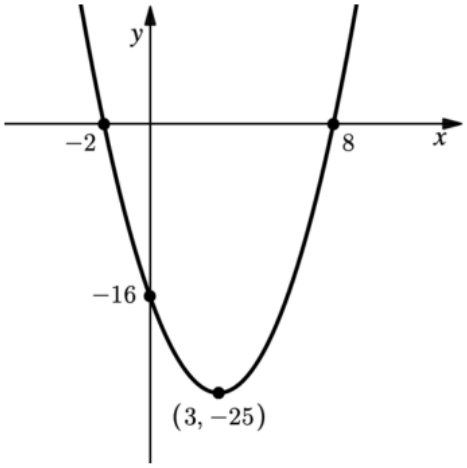


15 a)

$y = (x+2)(x-8)$

$\Rightarrow y = x^2 - 6x - 16$

$\Rightarrow y = (x-3)^2 - 25$

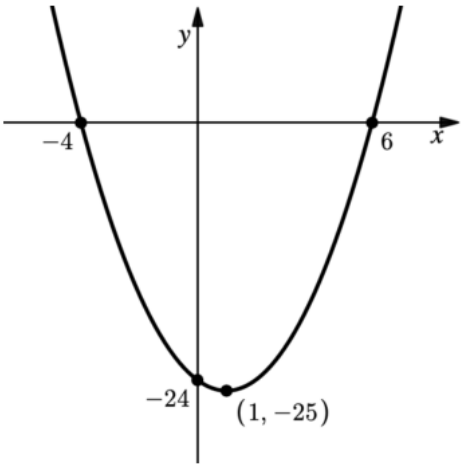


15 b)

$y = (x - 6)(x + 4)$

$y = x^2 - 2x - 24$

$y = (x - 1)^2 - 25$

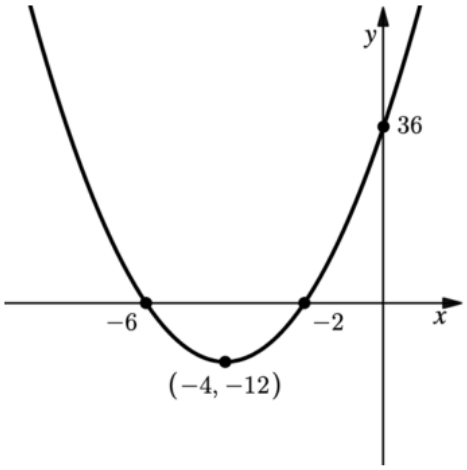


15 c)

$y = 3(x + 6)(x + 2)$

$y = 3x^2 + 24x + 36$

$y = 3(x + 4)^2 - 12$

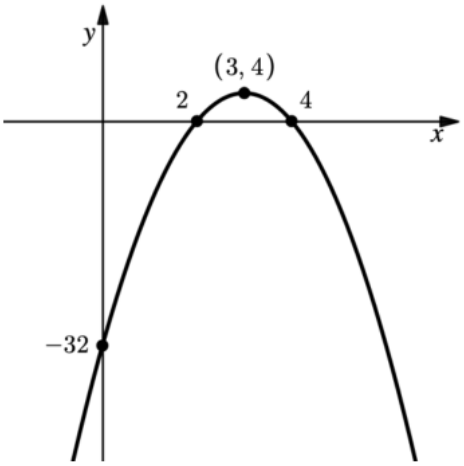


15 d)

$y = -4(x-2)(x-4)$

$y = -4x^2 + 24x - 32$

$y = -4(x-3)^2 + 4$

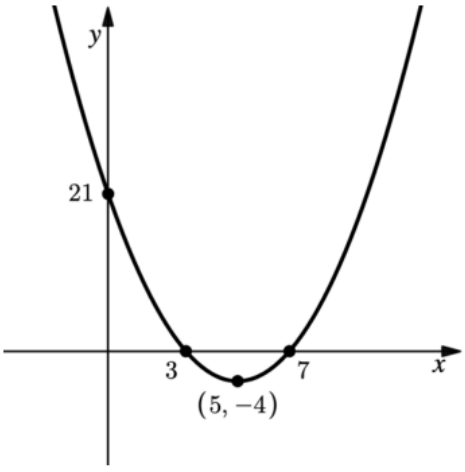


16 a)

$y = (x-3)(x-7)$

$y = x^2 - 10x + 21$

$y = (x-5)^2 - 4$

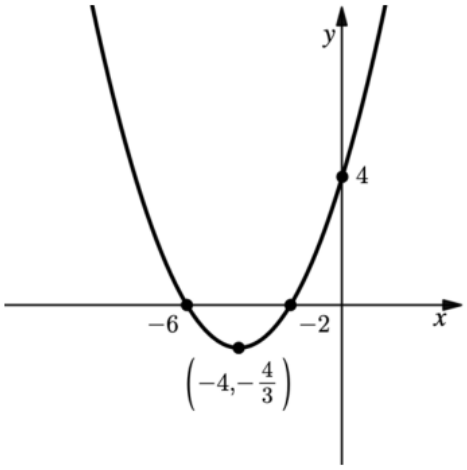


16 b)

$y = \frac{1}{3}(x+2)(x+6)$

$y = \frac{1}{3}x^2 + \frac{8}{3}x + 4$

$y = \frac{1}{3}(x+4)^2 - \frac{4}{3}$

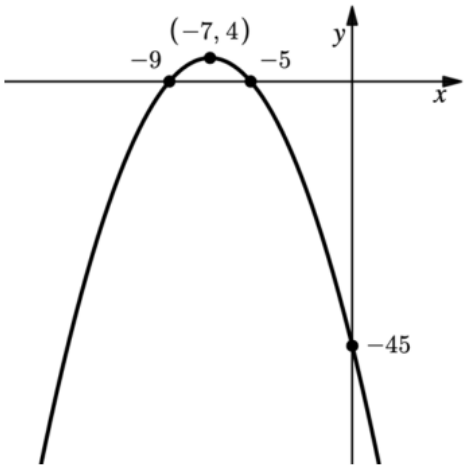


16 c)

$y = -(x+9)(x+5)$

$y = -x^2 - 14x - 45$

$y = -(x+7)^2 + 4$

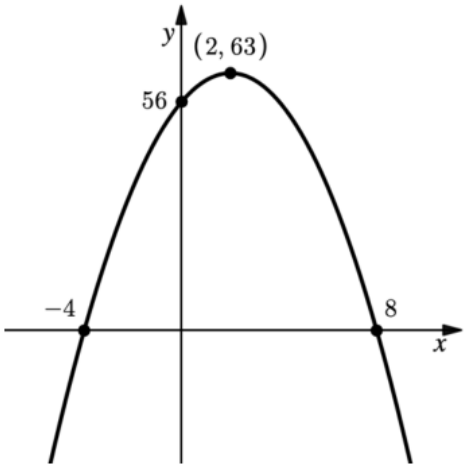


16 d)

$y = -\frac{7}{4}(x-8)(x+4)$

$y = -\frac{7}{4}x^2 + 7x + 56$

$y = -\frac{7}{4}(x-2)^2 + 63$

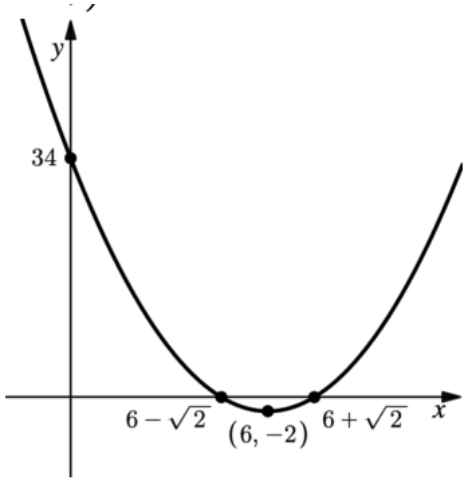


17 a)

$y = (x-6)^2 - 2$

$y = x^2 - 12x + 34$

$y = (x - (6-\sqrt{2}))(x - (6+\sqrt{2}))$

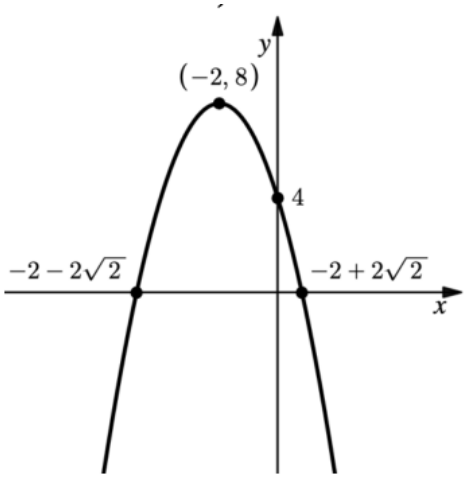


17 b)

$y = 8 - (x+2)^2$

$y = -x^2 - 4x + 4$

$y = -(x - (-2 - 2\sqrt{2}))(x - (-2 + 2\sqrt{2}))$

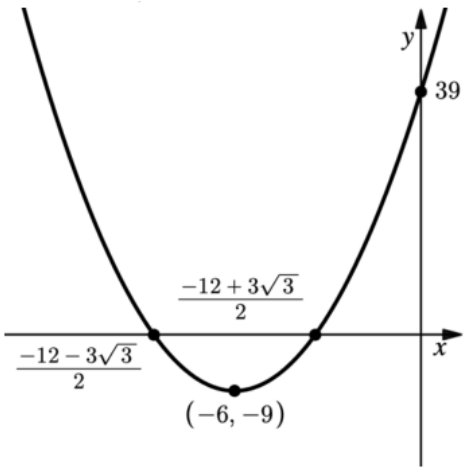


17 c)

$y = \frac{4}{3}(x+6)^2 - 9$

$y = \frac{4}{3}x^2 + 16x + 39$

$y = \frac{4}{3}(x - (\frac{-12-3\sqrt{3}}{2}))(x - (\frac{-12+3\sqrt{3}}{2}))$

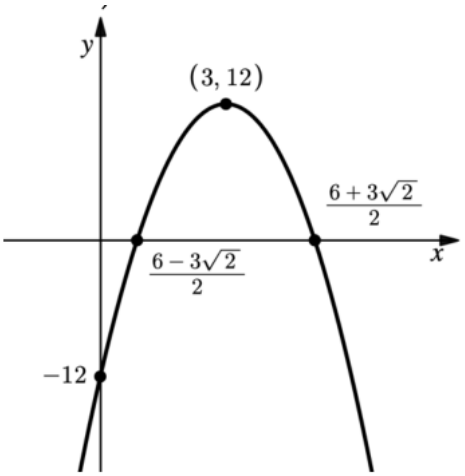


17 d)

$y = -\frac{8}{3}(x-3)^2 + 12$

$y = -\frac{8}{3}x^2 + 16x - 12$

$y = -\frac{8}{3}(x - (\frac{6-3\sqrt{2}}{2}))(x - (\frac{6+3\sqrt{2}}{2}))$

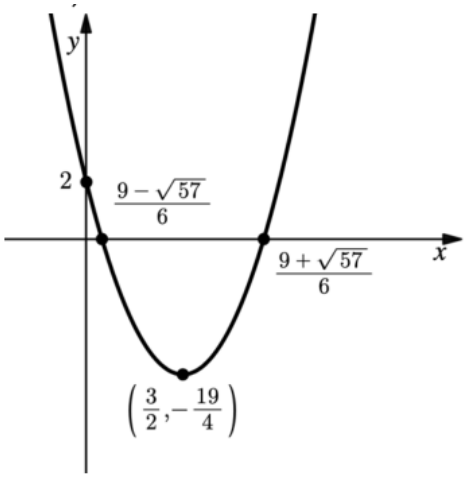


18 a)

$y = 3x^2 - 9x + 2$

$y = 3(x - \frac{3}{2})^2 - \frac{19}{4}$

$y = 3(x - (\frac{9+\sqrt{57}}{6}))(x - (\frac{9-\sqrt{57}}{6}))$

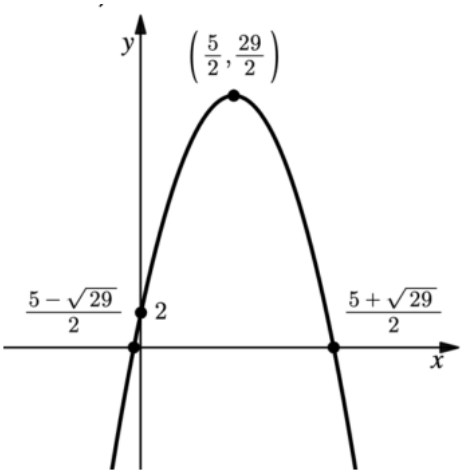


18 b)

$y = -2x^2 + 10x + 2$

$y = -2(x - \frac{5}{2})^2 + \frac{29}{2}$

$y = -2(x - (\frac{5+\sqrt{29}}{2}))(x - (\frac{5-\sqrt{29}}{2}))$

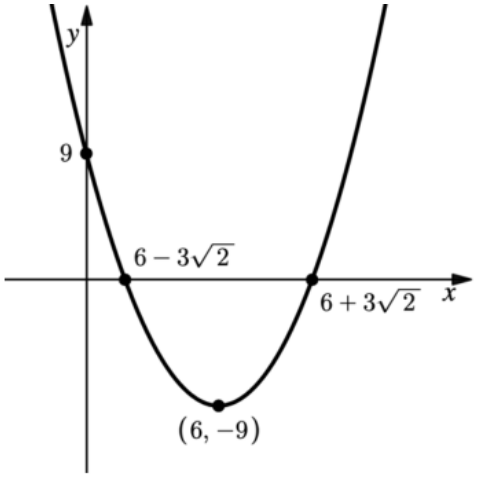


18 c)

$y = \frac{1}{2}x^2 - 6x + 9$

$y = \frac{1}{2}(x - 6)^2 - 9$

$y = \frac{1}{2}(x - (6 + 3\sqrt{2}))(x - (6 - 3\sqrt{2}))$

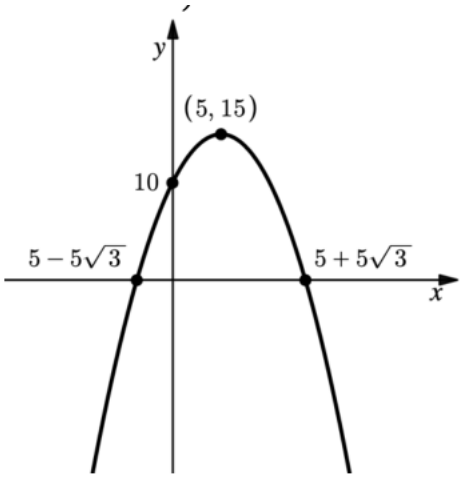


18 d)

$y = -\frac{1}{5}x^2 + 2x + 10$

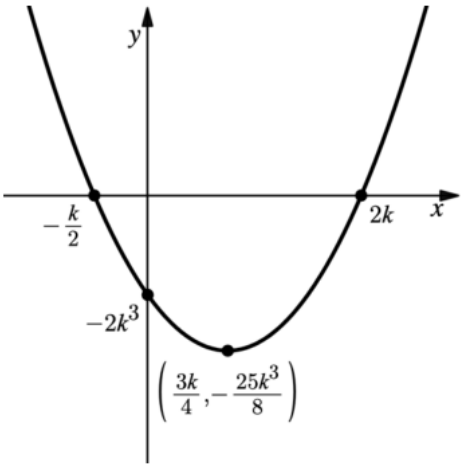
$y = -\frac{1}{5}(x-5)^2 + 15$

$y = -\frac{1}{5}(x-(5+5\sqrt{3}))(x-(5-5\sqrt{3}))$

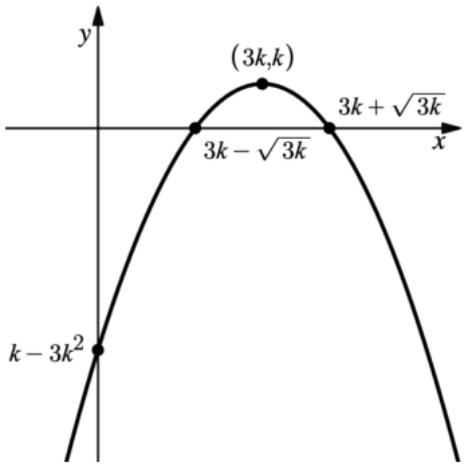


19 a)

$$
\begin{aligned}
& y = k(2x+k)(x-2k) \\
\Rightarrow\ & y = k(2x^2 - 3kx - 2k^2) \\
\Rightarrow\ & y = 2kx^2 - 3k^2x - 2k^3 \\
\Rightarrow\ & y = 2k(x^2 - \frac{3k}{2}x) - 2k^3 \\
\Rightarrow\ & y = 2k((x - \frac{3k}{4})^2 - \frac{9k^2}{16}) - 2k^3 \\
\Rightarrow\ & y = 2k(x - \frac{3k}{4})^2 - \frac{9k^3}{8} - 2k^3 \\
\Rightarrow\ & y = 2k(x - \frac{3k}{4})^2 - \frac{25k^3}{8}
\end{aligned}
$$



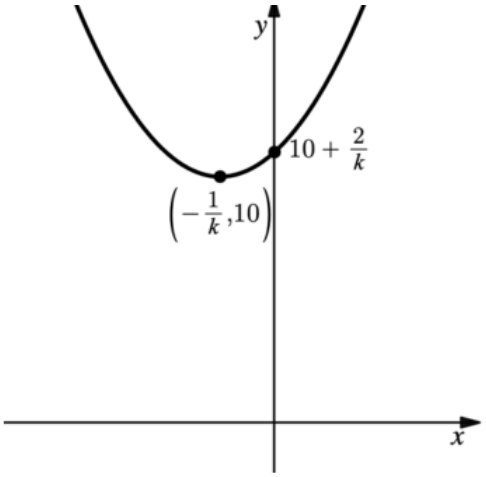
19 b)



19 c)

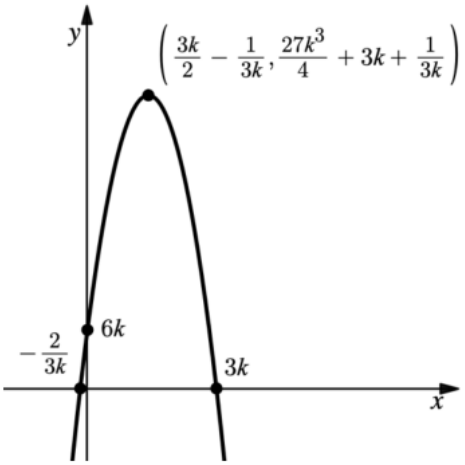
$y = 10 + \frac{2}{k}(kx+1)^2 \Rightarrow y = 10 + \frac{2}{k}(k(x+\frac{1}{k}))^2$

$\Rightarrow y = 10 + \frac{2}{k} \times k^2(x+\frac{1}{k})^2 \Rightarrow y = 10 + 2k(x+\frac{1}{k})^2$



19 d)

$$
\begin{aligned}
& y = (3kx + 2)(3k - x) \\
\Rightarrow\ & y = 9k^2x - 3kx^2 + 6k - 2x \\
\Rightarrow\ & y = -3kx^2 + (9k^2 - 2)x + 6k \\
\Rightarrow\ & y = -3k(x^2 - (3k - \frac{2}{3k})x) + 6k \\
\Rightarrow\ & y = -3k((x - (\frac{3k}{2} - \frac{1}{3k}))^2 - (\frac{3k}{2} - \frac{1}{3k})^2) + 6k \\
\Rightarrow\ & y = -3k((x - (\frac{3k}{2} - \frac{1}{3k}))^2 - (\frac{9k^2}{4} - 1 + \frac{1}{9k^2})) + 6k \\
\Rightarrow\ & y = -3k(x - (\frac{3k}{2} - \frac{1}{3k}))^2 + \frac{27k^3}{4} - 3k + \frac{1}{3k} + 6k \\
\Rightarrow\ & y = -3k(x - (\frac{3k}{2} - \frac{1}{3k}))^2 + \frac{27k^3}{4} + 3k + \frac{1}{3k}
\end{aligned}
$$



20)

$(k-x)(3x+k)=k-x \Rightarrow 3x+k=1 \Rightarrow x=\frac{1-k}{3}$

$\therefore y=k-\frac{1-k}{3} \Rightarrow y=k-\frac{1}{3}+\frac{k}{3} \Rightarrow y=\frac{4k}{3}-\frac{1}{3} \Rightarrow y=\frac{4k-1}{3}$

